ChatGPT

Probability in Normal Distribution

2/11/2026, 12:08:08 AM

Source: https://chatgpt.com/c/698c0872-463c-832b-8e64-1155a1ede123#

For a large population, the probability of a random variable falls above 2; + 20s

It looks like your question is not fully clear, but I think you are asking something like:

"For a large population, what is the probability that a random variable is greater than 2 (i.e., P(Z > 2))?"

If we are talking about a **standard normal distribution (Z-distribution)**, then:

$$P(Z > 2)$$

From the Z-table:

$$P(Z < 2) = 0.9772$$

So,

$$P(Z > 2) = 1 - 0.9772 = 0.0228$$

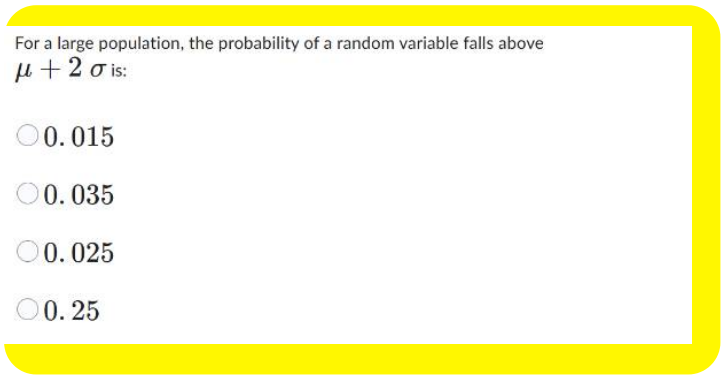
Final Answer:

The probability that a random variable is greater than 2 is:

$$\boxed{0.0228}$$

This means only **2.28%** of the values lie above 2 in a standard normal distribution.

If this is not exactly what you meant (for example, if you meant something else by "+ 20s"), please send the full question and I will explain in simple English.



The question is:

For a large population, the probability that a random variable falls above $\mu + 2\sigma$ is:

If the population is **normally distributed**, then:

$$\mu + 2\sigma$$

means **2 standard deviations above the mean**.

Now convert to Z-score:

$$Z = \frac{X - \mu}{\sigma}$$

For $X = \mu + 2\sigma$:

$$Z = 2$$

So we need:

$$P(Z > 2)$$

From the standard normal table:

$$P(Z > 2) = 0.0228$$

This is approximately **0.025**.

✅ Correct Answer:

0.025

(Because 0.0228 ≈ 0.025)

Simple understanding:
About **2.5% of the data** lies above $\mu + 2\sigma$ in a normal distribution.

For a large population, the probability of a random variable falls below the mean is 50%.

Question 2 options:
True
False explain properly

Answer: False.

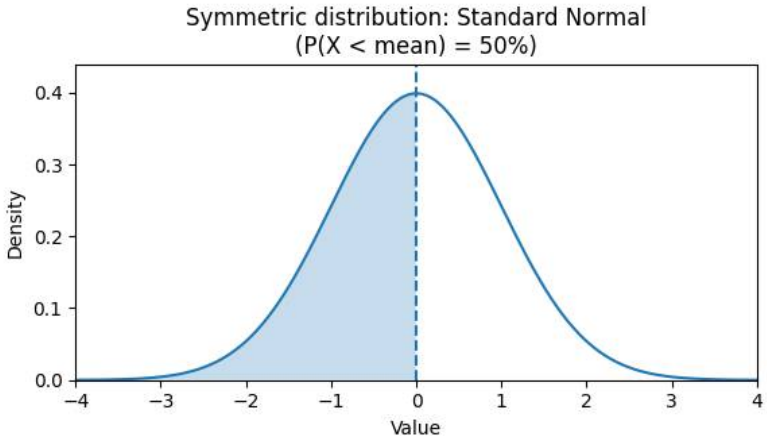
Explanation (simple English):

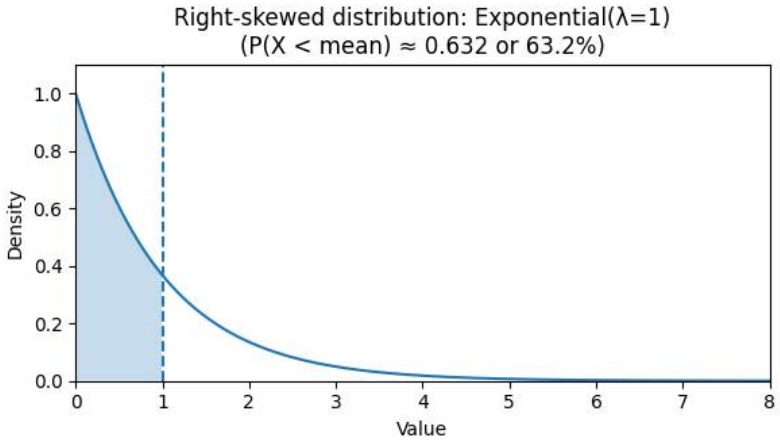
- The mean is the *average* value, not always the middle value.
- The statement "probability falls below the mean is 50%" is **only true** when the distribution is **symmetric** about the mean (for example, the normal distribution). In that case the mean = median and exactly half the values lie below the mean.
- For many real distributions the mean ≠ median. If the distribution is **right-skewed** (long tail to the right), the mean is pulled to the right of the median, so **more than 50%** of values lie below the mean.
  Example: Exponential distribution with mean = 1 → $P(X < 1) = 1 - e^{-1} \approx 0.632$ (63.2%).
- If the distribution is **left-skewed** (tail to the left), the mean is left of the median, so **less than 50%** of values lie below the mean.
  Simple discrete example: 90% of values are 10, and 10% are 0. Mean = 9. Probability $X < 9$ = probability of value 0 = 0.1 (10%), which is much less than 50%.
- Also, the phrase "large population" does **not** change this fact — it depends on the shape of the distribution, not the population size.

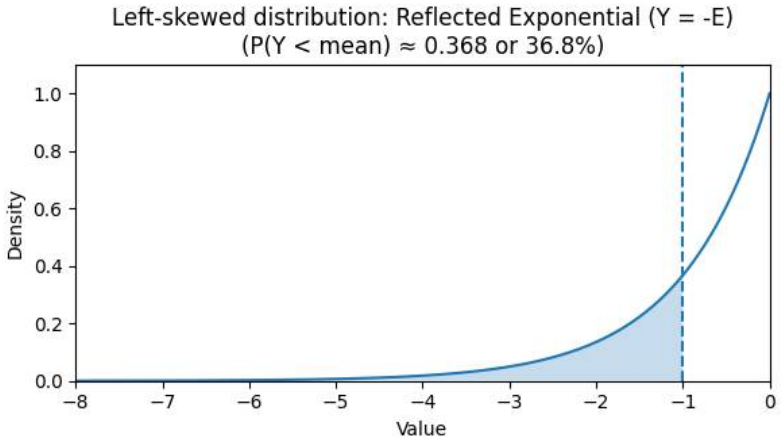
Takeaway: Only when the distribution is symmetric (mean = median) will exactly 50% of observations fall below the mean. Otherwise it can be less or more than 50%.

* The statement “probability falls below the mean is 50%” is **only true** when the distribution is **symmetric** about the mean (for example, the normal distribution). In that case the mean = median and exactly half the values lie below the mean. * For many real distributions the mean ≠ median. If the distribution is **right-skewed** (long tail to the right), the mean is pulled to the right of the median, so **more than 50%** of values lie below the mean. _Example:_ Exponential distribution with mean = 1 → P(X<1)\=1−e−1≈0.632P(X<1)=1-e^{-1}\\approx0.632P(X<1)\=1−e−1≈0.632 (63.2%). * If the distribution is **left-skewed** (tail to the left), the mean is left of the median, so **less than 50%** of values lie below the mean. _Simple discrete example:_ 90% of values are 10, and 10% are 0. Mean = 9. Probability X<9X<9X<9 = probability of value 0 = 0.1 (10%), which is much less than 50%. * Also, the phrase “large population” does **not** change this fact — it depends on the shape of the distribution, not the population size.

explain me properly with graph and in simple words







Nice — the three graphs above show the three cases in simple words. Below I explain each one and why the probability of being *below the mean* can be 50%, more than 50%, or less than 50%.

1) Symmetric distribution (Standard Normal)

- What you saw: a bell-shaped curve with the mean (vertical dashed line) exactly in the centre, and the shaded area to the left of the mean.
- What it means in simple words: the curve is the same on both sides of the mean, so exactly half of the values are on the left and half on the right.
- Probability below mean = **50%**.

2) Right-skewed distribution (Exponential)

- What you saw: a curve that starts high at zero and has a long tail to the right. The mean (dashed line) is pulled to the right by the long tail.
- What it means in simple words: because the mean is pulled right, **more than half** of the values lie left of the mean (they are the "bulk" near zero), and only a smaller fraction are in the long right tail.
- Numerical example: for Exp(λ=1) the mean = 1, and $P(X < 1) = 1 - e^{-1} \approx 0.632$.
  So **about 63.2%** of values are below the mean.

3) Left-skewed distribution (Reflected exponential, Y = −E)

- What you saw: a mirror of the exponential, with a long tail to the left. The mean is pulled left by the tail.
- What it means in simple words: because the mean sits left of most of the bulk of values, **less than half** the values lie below the mean.
- Numerical example: for Y = −E (E~Exp(1)) mean = −1, and $P(Y < -1) = e^{-1} \approx 0.368$.
  So **about 36.8%** of values are below the mean.

Final takeaway (very simple)

- The statement "probability of a random variable falls below the mean is 50%" is **only always true** when the distribution is **symmetric** (like the normal).

- For skewed distributions, the mean shifts toward the tail:
  - Right-skew → **> 50%** below the mean.
  - Left-skew → **< 50%** below the mean.

If you want, I can:

- show the median on the same plots so you can see mean vs median, or
- make a small animation or show sample histograms to visualise with actual random draws. Which would you prefer?